

1 核心交换图（逐步版）

为使交换结构更清晰，先约定两个记号：

$\mathbb{C}_{\text{can}}^N :=$ 大环 canonical slot 空间, $\mathbb{C}_{\text{coeff}}^N :=$ 大环 coefficient-packed slot 空间.

类似地,

$$\mathbb{C}_{\text{can}}^n, \quad \mathbb{C}_{\text{coeff}}^n$$

分别表示小环的 canonical slot 空间与 coefficient-packed slot 空间。

再记

$$S = \bigoplus_{r=0}^{k-1} S_r, \quad \text{EvalMod}_N = f^{\oplus N}, \quad \text{EvalMod}_n = f^{\oplus n}.$$

Step 1: C2S 与 Split_k 的交换图

$$\begin{array}{ccc} \mathbb{C}_{\text{can}}^N & \xrightarrow{\text{Split}_k} & \bigoplus_{r=0}^{k-1} \mathbb{C}_{\text{can}}^n \\ \downarrow \text{C2S}_N & & \downarrow \bigoplus_{r=0}^{k-1} \text{C2S}_n \\ \mathbb{C}_{\text{coeff}}^N & \xrightarrow{S} & \bigoplus_{r=0}^{k-1} \mathbb{C}_{\text{coeff}}^n \end{array}$$

对应关系为

$$\left(\bigoplus_{r=0}^{k-1} \text{C2S}_n \right) \circ \text{Split}_k = S \circ \text{C2S}_N.$$

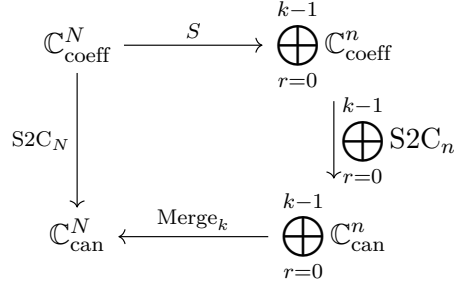
Step 2: EvalMod 与 coefficient 分组的交换图

$$\begin{array}{ccc} \mathbb{C}_{\text{coeff}}^N & \xrightarrow{S} & \bigoplus_{r=0}^{k-1} \mathbb{C}_{\text{coeff}}^n \\ \downarrow f^{\oplus N} & & \downarrow \bigoplus_{r=0}^{k-1} f^{\oplus n} \\ \mathbb{C}_{\text{coeff}}^N & \xrightarrow{S} & \bigoplus_{r=0}^{k-1} \mathbb{C}_{\text{coeff}}^n \end{array}$$

对应关系为

$$\left(\bigoplus_{r=0}^{k-1} \text{EvalMod}_n \right) \circ S = S \circ \text{EvalMod}_N.$$

Step 3: S2C 与 Merge_k 的交换图

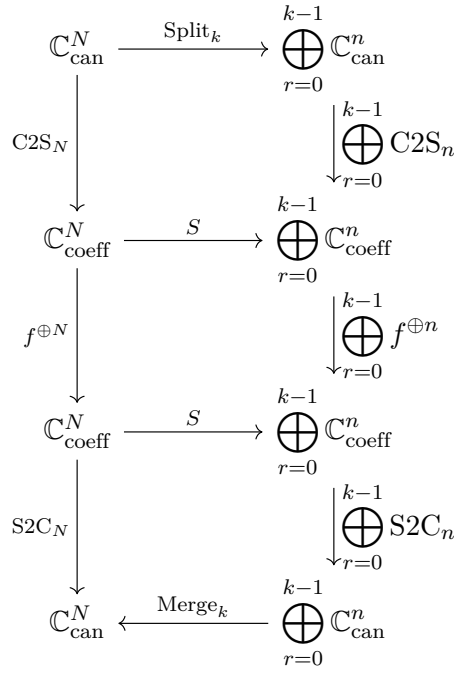


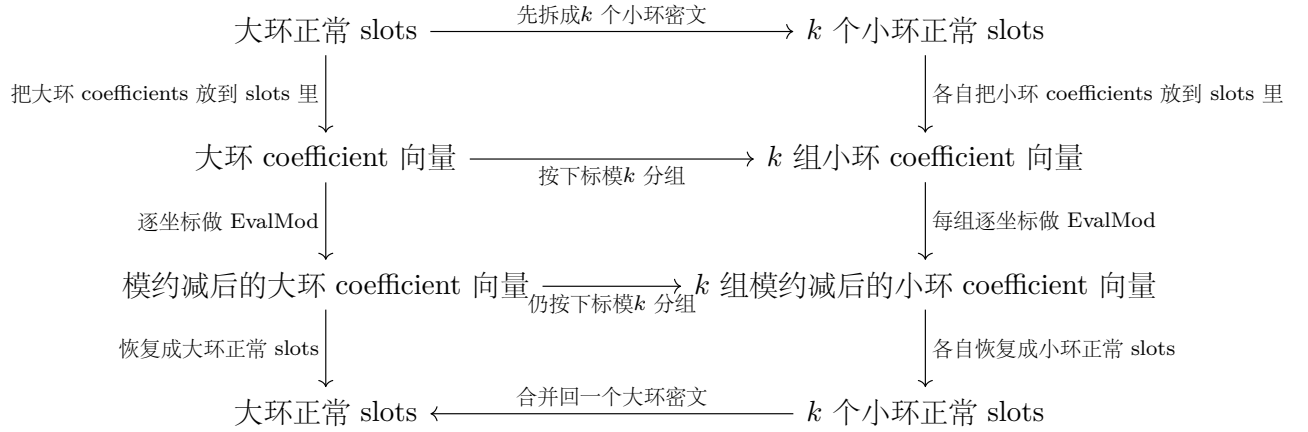
对应关系为

$$\text{Merge}_k \circ \left(\bigoplus_{r=0}^{k-1} \text{S2C}_n \right) \circ S = \text{S2C}_N .$$

Step 4: 将三步串起来的整体交换图

把 Step 1, Step 2, Step 3 纵向串联，可得完整 bootstrapping core 分解图：





因此整体满足

$$\text{Merge}_k \circ \left(\bigoplus_{r=0}^{k-1} (\text{S2C}_n \circ \text{EvalMod}_n \circ \text{C2S}_n) \right) \circ \text{Split}_k = \text{S2C}_N \circ \text{EvalMod}_N \circ \text{C2S}_N.$$

若记

$$\text{BTS}_N := \text{S2C}_N \circ \text{EvalMod}_N \circ \text{C2S}_N, \quad \text{BTS}_n := \text{S2C}_n \circ \text{EvalMod}_n \circ \text{C2S}_n,$$

则上式可简写为

$$\text{Merge}_k \circ \left(\bigoplus_{r=0}^{k-1} \text{BTS}_n \right) \circ \text{Split}_k = \text{BTS}_N.$$

Step 5: 考虑实现中的全局归一化因子

若真实实现中存在全局常数 γ_k ，则更保守地写为

$$\text{Merge}_k^{\text{impl}} \circ \left(\bigoplus_{r=0}^{k-1} \text{BTS}_n^{\text{impl}} \right) \circ \text{Split}_k^{\text{impl}} \approx \gamma_k \text{BTS}_N^{\text{impl}}.$$

等价地，

$$\gamma_k^{-1} \text{Merge}_k^{\text{impl}} \circ \left(\bigoplus_{r=0}^{k-1} \text{BTS}_n^{\text{impl}} \right) \circ \text{Split}_k^{\text{impl}} \approx \text{BTS}_N^{\text{impl}}.$$